

Course Handout

Institute/School Name	Chitkara University Institute of Engineering and Technology		
Department Name	Department of Computer Science & Engineering		
Programme Name	B.E (Computer Science & Engineering)		
Course Name	Numerical Aptitude & Logical Reasoning-I	Session	July-Dec 2026
Course Code	25UNI0110	Semester/Batch	5 th /2024
L-T-P-C	3-0-0	Course Credits	03
Pre-requisite	NA	NHEQF Level¹	5.5
Course Coordinator	Dr. Himanshu Gupta	SDG Number⁴	4, 5, 8, 15

1. Objectives of the Course

- To develop a strong foundation in numerical concepts and arithmetic operations for solving quantitative aptitude problems.
- To enhance logical and analytical reasoning skills through relation-based, coding–decoding, syllogism, and pattern-identification problems.
- To build proficiency in applied arithmetic, including percentages, interest, profit & loss, time & work, and progression-based calculations.
- To improve speed and accuracy in solving sequence, series, calendar, clock, and time-based reasoning questions.
- To strengthen verbal aptitude skills in reading comprehension, vocabulary usage, sentence arrangement, and grammatical error correction.

2. Course Learning Outcomes (CLOs)

After completion of the course, the student should be able to:

	Course Learning Outcome	*POs	NHEQF Level Descriptor ²	Sessions
CLO 01	Apply fundamental numerical and arithmetic concepts to solve problems related to number systems, ratios, percentages, and basic finance.	PO1, PO5, PO6, PO8, PSO1	• Apply standard quantitative and analytical methods to solve numerical and data interpretation problems.	1-12
CLO 02	Analyze and solve logical reasoning and pattern-based problems involving relations, coding–decoding, directions, syllogisms, and sequences.	PO1, PO5, PO6, PO8, PO10	• Analyze and solve logical reasoning problems using structured approaches.	13-24
CLO 03	Solve applied arithmetic problems related to time, work, profit–loss, interest, and progression-based calculations with accuracy.	PO5, PO6, PO8, PO11	• Interpret and evaluate graphical and non-verbal data to draw logical conclusions.	25-36
CLO 04	Demonstrate verbal aptitude skills through effective reading comprehension, vocabulary usage, sentence rearrangement, and grammatical correction.	PO1, PO2, PO3, PO5, PO6, PO8, PSO2	• Use critical thinking and judgment to make decisions in complex reasoning scenarios.	37-48
CLO 05	Evaluate and solve aptitude-based problems by selecting appropriate logical, quantitative, and verbal reasoning techniques to enhance competitive examination and placement readiness.	PO4, PO5, PO7, PO9	• Apply rule-based correction and improvement techniques.	49-60
Total Contact Hours				60

CLO-PO Mapping

CLO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	Type of Assessment's ³
CLO01	L	-	-	-	L	M	-	H	-	-	-	H	-	Formative/ Summative
CLO02	L	-	-	-	M	H	-	H	-	H	-	-	-	Formative/ Summative
CLO03	-	-	-	-	L	M	-	H	-	-	H	-	-	Formative/ Summative
CLO04	H	H	H	-	H	M	-	H	-	-	-	-	H	Formative/ Summative
CLO05	-	-	-	H	H	-	M	-	H	-	-	-	-	Formative/ Summative

H=High, M=Medium, L=Low

¹ National Higher Education Qualification Framework Level, Refer to annexure

² NHEQF Level Descriptor, Refer to Annexure & [Learning outcomes descriptors for qualification for all levels on the NHEQF](#)

³ Types of Assessments can be referred from Type of Assessments. Refer to Annexure.

⁴ For SDG Mapping with Courses, Pl refer [SDG Mapping policy for Courses](#)

3. Recommended Books:

B01: Quantitative Aptitude for Competitive Examinations — *Dr. R.S. Aggarwal* (Comprehensive coverage of number system, HCF/LCM, averages, percentages, ratios, profit-loss, time & work, basic finance and more)

B02: Fast Track Objective Arithmetic — *Rajesh Verma* (Clear concepts, shortcuts, and graded practice for arithmetic topics)

B03: A Modern Approach to Logical Reasoning — *Dr. R.S. Aggarwal* (Reasoning fundamentals including directions, blood relation, coding-decoding, syllogism)

Textbooks:

B04: Analytical Reasoning — *M.K. Pandey* (In-depth practice and techniques for reasoning and logic)

B05: Objective General English — *S.P. Bakshi* (Grammar, sentence structure, error-spotting) — widely used for competitive preparation. (general recommendation, common resource)

4. Other readings and relevant websites:

Serial No	Link of Journals, Magazines, websites and Research Papers
1.	https://www.mdpi.com/2227-7102/15/3/262
2.	https://www.geeksforgeeks.org/aptitude/verbal-ability/
3.	https://en.wikipedia.org/wiki/Reading_Research_Quarterly
4.	https://ejournals.org/wp-content/uploads/Impact-of-Students-Vocabulary-Level.pdf

5. Recommended Tools and Platforms: NA

6. Course Plan:

Lecture No. (Hour)	Topic(s)	Textbook
1-10	Module-1: Foundational Logic & Number Concepts 1. Blood Relation 2. Coded Relation 3. Analogy 4. Direction 5. Number System 6. H.C.F. & L.C.M. 7. Average 8. Remainder Theorem	B04, B05
11-20	Module-2: Ratio, Logic & Group Concepts 1. Ratio & Proportion 2. Ages 3. Partnership 4. Allegation 5. Odd Man Out 6. Syllogism	B04, B05
ST-I		
21-29	Module-3: Calculation, Pattern & Basic Finance 1. Percentage 2. Alphabet Test 3. Mathematical Operation 4. Coding-Decoding 5. Simple Interest	B04, B05
30-38	Module-4: Applied Arithmetic & Sequences 1. Compound Interest 2. Profit & Loss 3. Discount 4. Number Sequence 5. Series 6. Arithmetic Progression (A.P) & Geometric Progression (G.P)	B04, B05
ST-II		

39-50	Module-5: Time, Work & Spatial Logic 1. Time & Work 2. Pipe & Cistern 3. Mixed Proportion 4. Work & Wages 5. Ranking 6. Cube & Cuboid 7. Calendar 8. Clock 9. Time Sequence	B04, B05
51-60	Module-6: Verbal Aptitude 1. Reading Comprehension 2. Rearrangement/ Jumbled Sentences 3. Cloze Test 4. Vocabulary a. Synonyms & Antonyms b. Idioms & Phrases c. One-word Substitution 5. Error Detection & Sentence Improvement: a. Noun/Article/Verb/Subject-Verb Agreement	B04, B05
End Term: MCQs (Testpad)		

7. Delivery/Instructional Resources

S. No.	Syllabus	CLO	Book No, CH No., Page No.	TLM ³	ALM ⁴	Web References	Audio-Video
1	Module-1: Foundational Logic & Number Concepts 1. Blood Relation 2. Coded Relation 3. Analogy 4. Direction 5. Number System 6. H.C.F. & L.C.M. 7. Average 8. Remainder Theorem	CO1, CO2	B01, B02	Lecture, Discussion	Think/Pair / Share	https://www.mdpi.com/2227-7102/15/3/262	https://youtu.be/MiJn65MglTE?si=DQDT0VyO7hEVjUVI
2	Module-2: Ratio, Logic & Group Concepts 1. Ratio & Proportion 2. Ages 3. Partnership 4. Allegation 5. Odd Man Out 6. Syllogism	CLO2	B04, B05	Lecture, Discussion	Just in Time Teaching	https://www.geeksforgeeks.org/aptitude/verbal-ability/	https://youtu.be/Xsg6-1c-IGI?si=v00TLHEEiRB_wfEj
ST-I							
3	Module-3: Calculation, Pattern & Basic Finance 1. Percentage 2. Alphabet Test 3. Mathematical Operation 4. Coding-Decoding 5. Simple Interest	CO1, CO2	B01, B03, B04	Lecture, Discussion	Just in Time Teaching	https://www.mdpi.com/2227-7102/15/3/262	https://www.youtube.com/live/qVcEJxgJOQs?si=TZcbfaLObmxAK7kU
4	Module-4: Applied Arithmetic & Sequences 1. Compound Interest 2. Profit & Loss 3. Discount 4. Number Sequence 5. Series 6. Arithmetic Progression (A.P) & Geometric Progression (G.P)	CLO	B03, B05	Lecture, Discussion	Just in Time Teaching	https://en.wikipedia.org/wiki/Reading_Research_Quarterly	https://youtu.be/OIKmdGAmgNU?si=GtDrbBMUJ5oeJ_HD
ST-II							
5	Module-5: Time, Work & Spatial Logic 1. Time & Work 2. Pipe & Cistern 3. Mixed Proportion 4. Work & Wages 5. Ranking 6. Cube & Cuboid	CO1, CO3,	B03, B04	Lecture, Discussion	Just in Time Teaching	https://ejournals.org/wp-content/uploads/Impact-of-Students-	https://youtu.be/KE7tQf9spPg?si=6ilLbqxovoRCg03t

	7. Calendar 8. Clock 9. Time Sequence					Vocabulary- Level.pdf	
6	Module-6: Verbal Aptitude 1. Reading Comprehension 2. Rearrangement/ Jumbled Sentences 3. Cloze Test 4. Vocabulary a. Synonyms & Antonyms b. Idioms & Phrases c. One-word Substitution 5. Error Detection & Sentence Improvement: a. Noun/Article/Verb/Subject-Verb Agreement	CO4, CLO5	B03, B04	Lecture, Discussion	Just in Time Teaching	https://www.geeksforgeeks.org/aptitude/verbal-ability/	https://youtu.be/jj7Z-QZoiuQ?si=6sLPJZR3W5wjklJ6

8. Remedial Classes⁵:

Slow Learners	Average Learners	Fast Learners
Encouragement for improvement using peer Tutoring.	Doubt Session according to the slot free in timetable.	Revision on topics beyond those covered in CHO.

9. Self-Learning⁶:

S. No.	Topics	CLO	ALM	References/MOOCs
1	Puzzles, DI, Verbal Reasoning	CLO1, CLO2, CLO3, CLO4, CLO5	Focused Listening	https://www.geeksforgeeks.org/aptitude/verbal-ability/

10. Delivery Details of Content Beyond Syllabus⁷:

S. No.	Advanced Topics, Additional Reading, Research papers and any	CLO	POs	ALM	References/MOOCs
1	Article, Adjective, Error Detection, Verb, Sentence Rearrangement, Unseen Passage	CLO1	PO1, PO4, PO5, PO7	Focused Listening	https://www.mdpi.com/2227-7102/15/3/262

11. Evaluation Scheme & Components:

Assessment Type ⁸	Evaluation Component ⁹	Type of Component ¹⁰	No. of Assessments ¹¹	% Weightage of Component	Max. Marks	Mode of Assessment	CLO
Formative	Internal	Formative Assessment	02	20%	20 Each	Online	CLO 1, CLO 2
Formative	Internal	Sessional Test	02	30%	30 Each	Online	CLO 2, CLO 3
Formative	External	End Term	01	50%	50	Online	CLO 3, CLO 4, CLO5
Total			100%				

* Out of Two FA's and ST's, Best 1 FA and 1 ST will be considered.

**As per Academic Guidelines, a minimum of 75% attendance is required to become eligible for appearing in the End Semester Examination.

⁵ Refer to Annexure

⁶ Refer to Annexure

⁷ Refer to Annexure

⁸ Refer to [Annexure 2 of NCrf](#)

⁹ Refer to Annexure

¹⁰Refer to Annexure

¹¹ Refer to Annexure


12. Syllabus of the Course:

Subject: Numerical Aptitude & Logical Reasoning-I			
Modules	Topic(s)	No. of Lectures	Weightage %
Foundational Logic & Number Concepts	1. Blood Relation 2. Coded Relation 3. Analogy 4. Direction 5. Number System 6. H.C.F. & L.C.M. 7. Average 8. Remainder Theorem	10 Hrs	17%
Ratio, Logic & Group Concepts	1. Ratio & Proportion 2. Ages 3. Partnership 4. Allegation 5. Odd Man Out 6. Syllogism	10 Hrs	17%
ST-I			
Calculation, Pattern & Basic Finance	1. Percentage 2. Alphabet Test 3. Mathematical Operation 4. Coding-Decoding 5. Simple Interest	9 Hrs	15%
Applied Arithmetic & Sequences	1. Compound Interest 2. Profit & Loss 3. Discount 4. Number Sequence 5. Series 6. Arithmetic Progression (A.P) & Geometric Progression (G.P)	9 Hrs	15%
ST-II			
Time, Work & Spatial Logic	1. Time & Work 2. Pipe & Cistern 3. Mixed Proportion 4. Work & Wages 5. Ranking 6. Cube & Cuboid 7. Calendar 8. Clock 9. Time Sequence	12 Hrs.	19%
Verbal Aptitude	1. Reading Comprehension 2. Rearrangement/ Jumbled Sentences 3. Cloze Test 4. Vocabulary a. Synonyms & Antonyms b. Idioms & Phrases c. One-word Substitution 5. Error Detection & Sentence Improvement: a. Noun/Article/Verb/Subject-Verb Agreement	10 Hrs.	17%
End Term: MCQs (Testpad)			

13. Academic Integrity Policy:

Education at Chitkara University builds on the principle that excellence requires freedom where Honesty and integrity are its prerequisites. Academic honesty in the advancement of knowledge requires that all students and Faculty respect the integrity of one another's work and recognize the importance of acknowledging and safeguarding intellectual property. Any breach of the same will be tantamount to severe academic penalties.

This Document is approved by:

Designation	Name	Signature
Course Coordinator	Dr. Himanshu Gupta	
Head-Academic Delivery	Dr. Susheela Hooda	
Dean	Dr. Rupali Gill	
Date (DD/MM/YYYY)	22/06/2026	